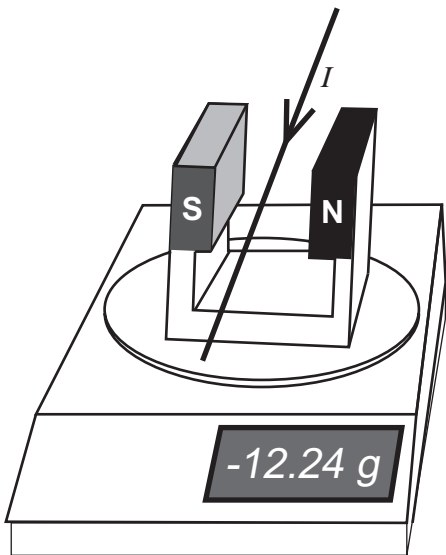


4. A student carries out an investigation. She uses a top pan balance to investigate the relationship $F=BI\ell$ using the set up shown in the diagram. The current-carrying wire is placed perpendicular to the magnetic field, B .

The current in the wire is varied using a variable power supply while the magnetic field, B , and the length of wire in the field, ℓ , are kept constant. Before switching on the current, the balance is set to zero.



- (a) (i) Explain why the reading on the balance is negative for the direction of current and magnetic field shown in the diagram. [3]

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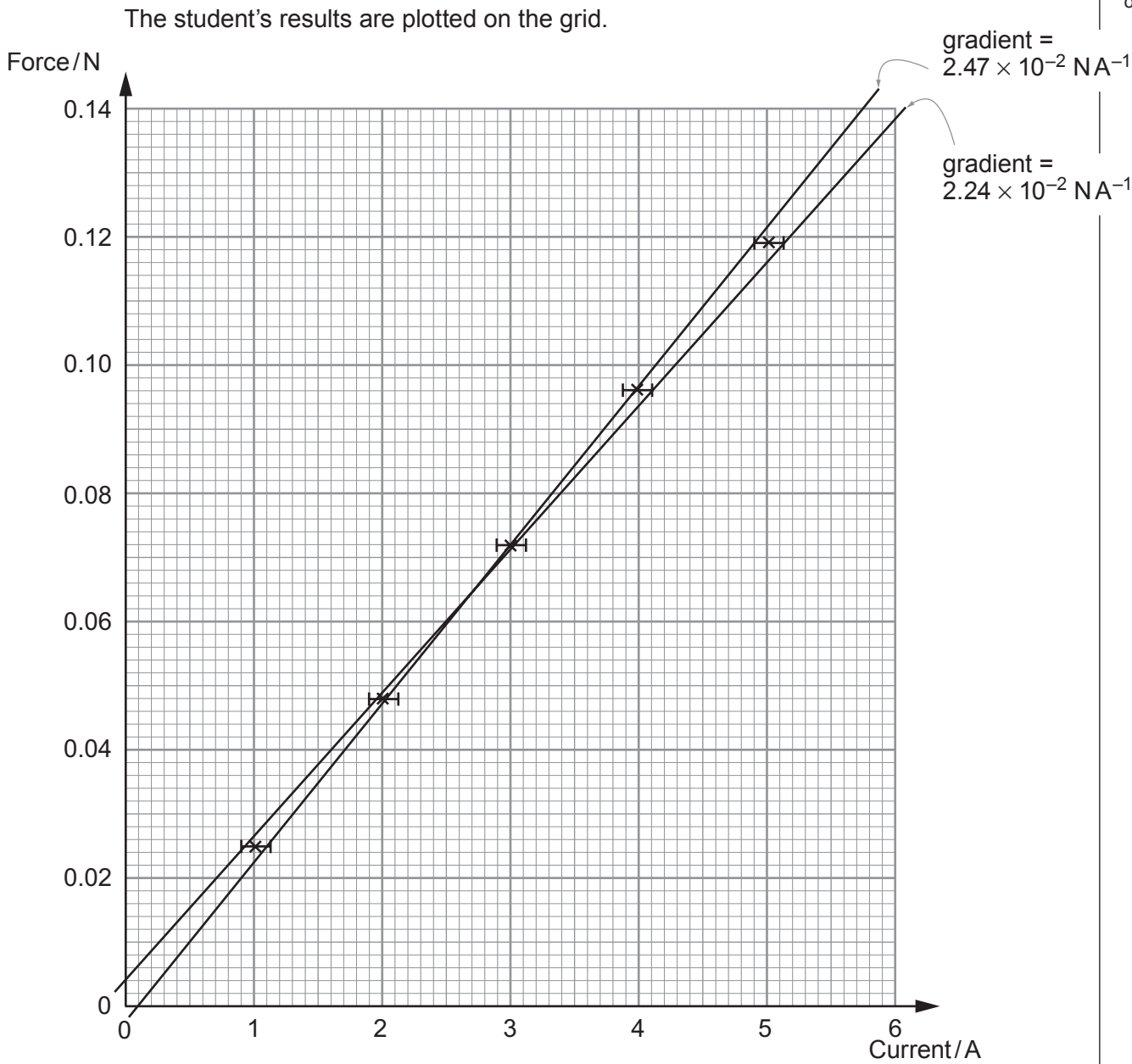
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- (ii) Calculate the mean value of the gradients ($2.47 \times 10^{-2} \text{ NA}^{-1}$ and $2.24 \times 10^{-2} \text{ NA}^{-1}$) and its **percentage** uncertainty. [3]

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- (iii) The length of wire in the uniform magnetic field is $(8.0 \pm 0.5)\text{ cm}$. Determine a value for the magnetic field strength and its **absolute** uncertainty, stating both to an appropriate number of significant figures. [4]

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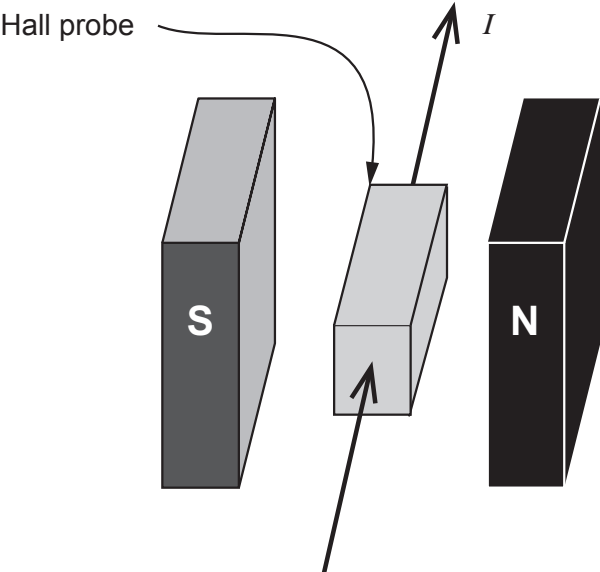
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- (b) In another experiment a Hall probe is used to confirm the value of the magnetic field strength obtained. The value of B obtained using the Hall probe is $(310 \pm 8)\text{ mT}$.



- (i) Explain how the Hall voltage arises in the Hall probe shown. [3]

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(ii) Discuss the effectiveness of the experiment in part (a) for confirming the relationship $F=BI\ell$ and measuring the magnetic field strength, compared with using the Hall probe {see (a)(iii) and (b)}. [6 QER]

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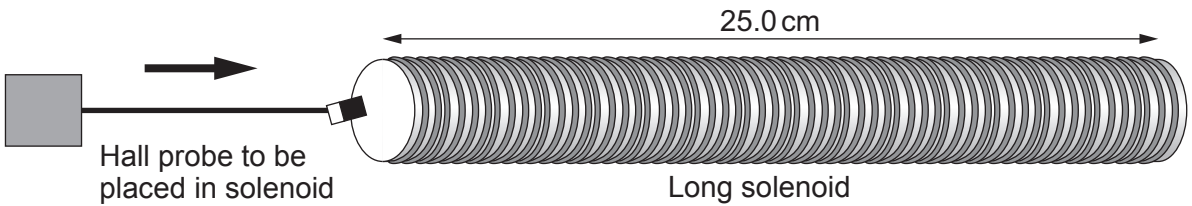
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4. An experiment is carried out to measure the magnetic field in a long solenoid as the current in the solenoid is varied.



Theory states that the relationship between magnetic flux density, B , at the centre of the solenoid and current, I , is given by the equation:

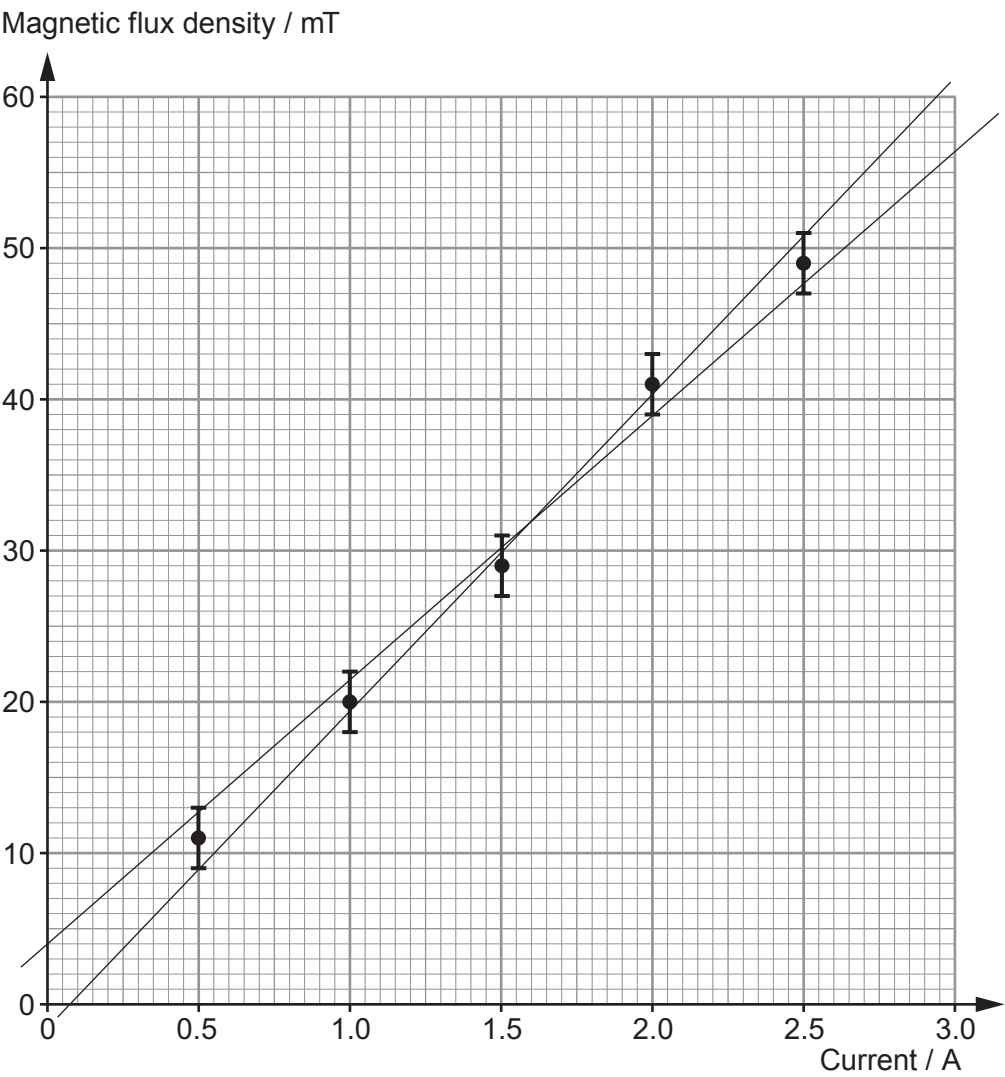
$$B = \mu_0 n I$$

The results obtained are shown in the table and plotted on the grid opposite along with error bars and lines of maximum and minimum gradient.

Current / A $\pm 0.01 \text{ A}$	Magnetic flux density / mT $\pm 2 \text{ mT}$
0.50	11
1.00	20
1.50	29
2.00	41
2.50	49



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(a) Calculate the mean value of the gradient along with its **percentage** uncertainty. [4]

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(b) Calculate the number of turns in the solenoid along with its **absolute** uncertainty. The percentage uncertainty in the length of the solenoid is 0.4%. [3]

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(c) (i) The solenoid manufacturer states that there are exactly 5000 turns in the solenoid. Evaluate the accuracy of your value obtained in part (b) and whether or not the graph is in agreement with the equation: $B = \mu_0 nI$. [4]

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(ii) Suggest a reason for the disagreement between the manufacturer's stated value (5000 turns) and your value calculated in part (b). Suggest how the experimental technique might be improved for better agreement. [2]

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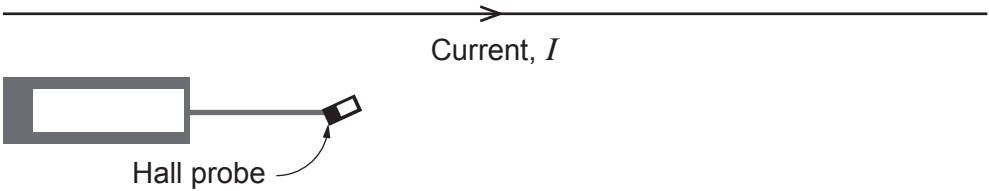
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4. (a) The current in a long wire is 290 A and is too great to be measured by an ammeter. Explain how you could use a Hall probe, calibrated in tesla (T), to determine this current. [3]



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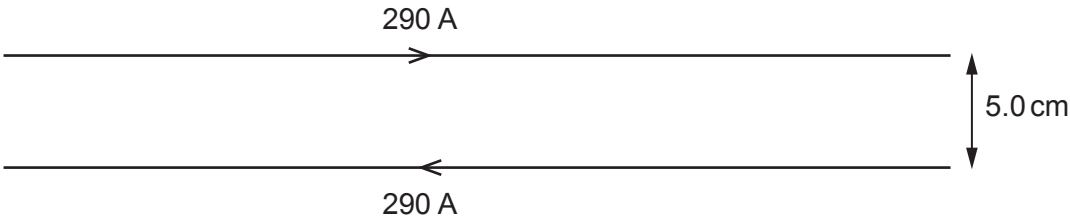
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- (b) Another long wire carrying the same large current is placed parallel to the original wire as shown. Calculate the force per unit length on each wire also stating the direction of the force on each wire. [4]



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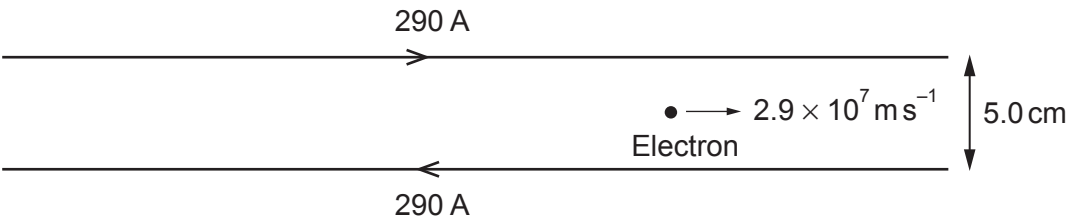
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- (c) (i) Tirion claims that an electron halfway between the wires, travelling at a speed of $2.9 \times 10^7 \text{ ms}^{-1}$ parallel to the wires will perform perfect circular motion between the wires. Determine, using a suitable calculation, whether or not Tirion's claim is correct. The magnetic flux density halfway between the wires is 4.64 mT . [4]



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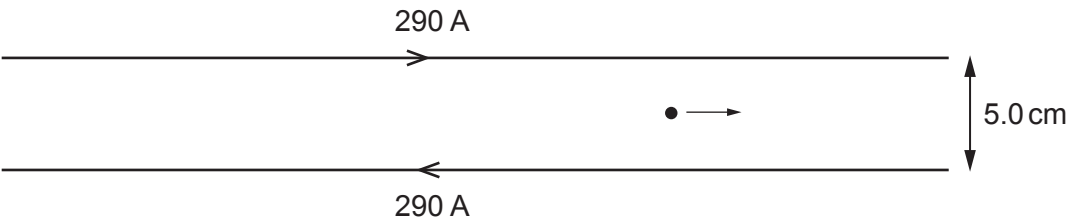
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- (ii) Sketch the motion of the electron. [2]

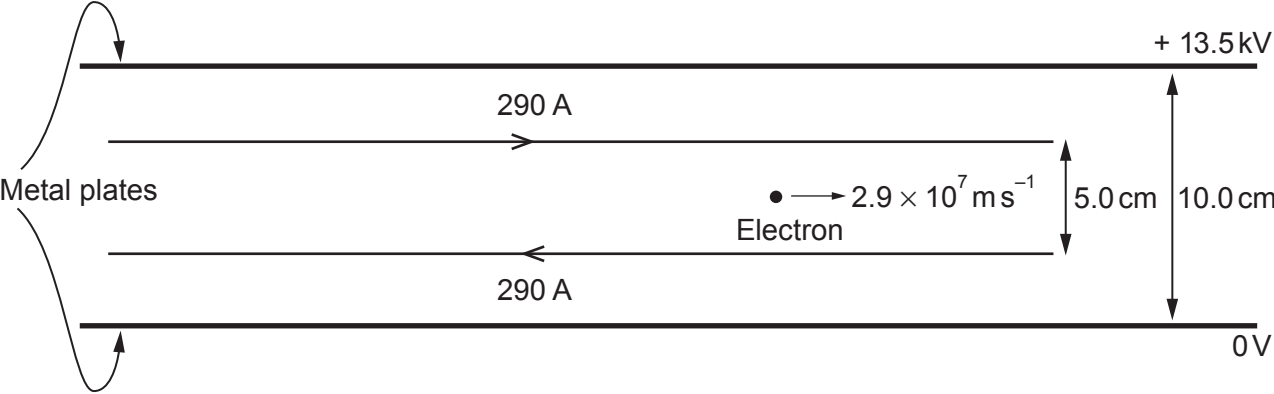


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(d) Suppose that an electron travels in a region with a magnetic field and an electric field due to two parallel metal plates as shown below. Deduce whether or not the electron continues with constant velocity ($B = 4.64 \text{ mT}$). [5]



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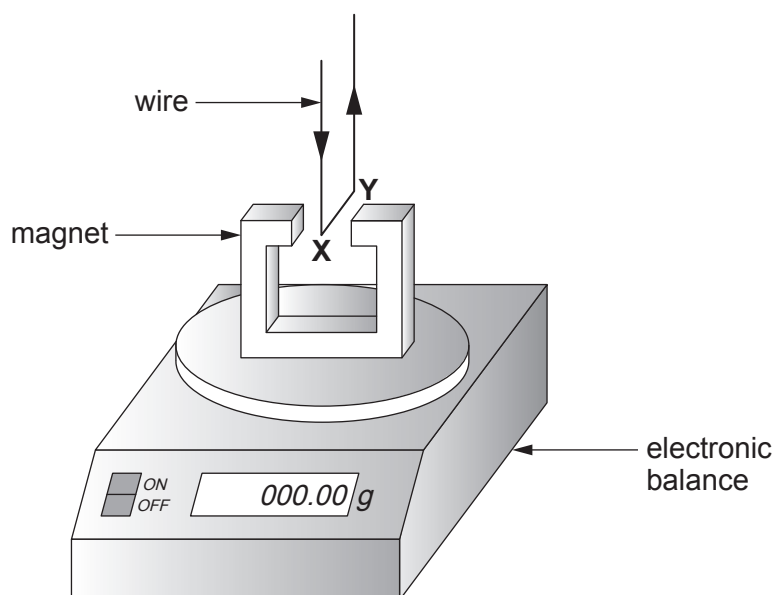
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4. The following set-up is used to investigate the force on a current-carrying wire in a magnetic field.



The part **XY** of the wire which is in the magnetic field, has been placed carefully so that the current is from front to back at 90° to the direction of the magnetic field. When there is no current the balance is reset to display 000.00 g. When there is a current in the direction shown, the magnetic force results in a positive reading on the display of the balance. However, this signifies an upward force on the wire.

- (a) (i) Explain briefly why the magnetic force must be upward on the part **XY** of the wire. [2]

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- (ii) **On the diagram**, indicate which is the North pole of the magnet, **and** state below which rule you used to obtain your answer. [2]

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- (iii) Explain why the vertical parts of the wire have no effect on the reading displayed by the electronic balance. [1]

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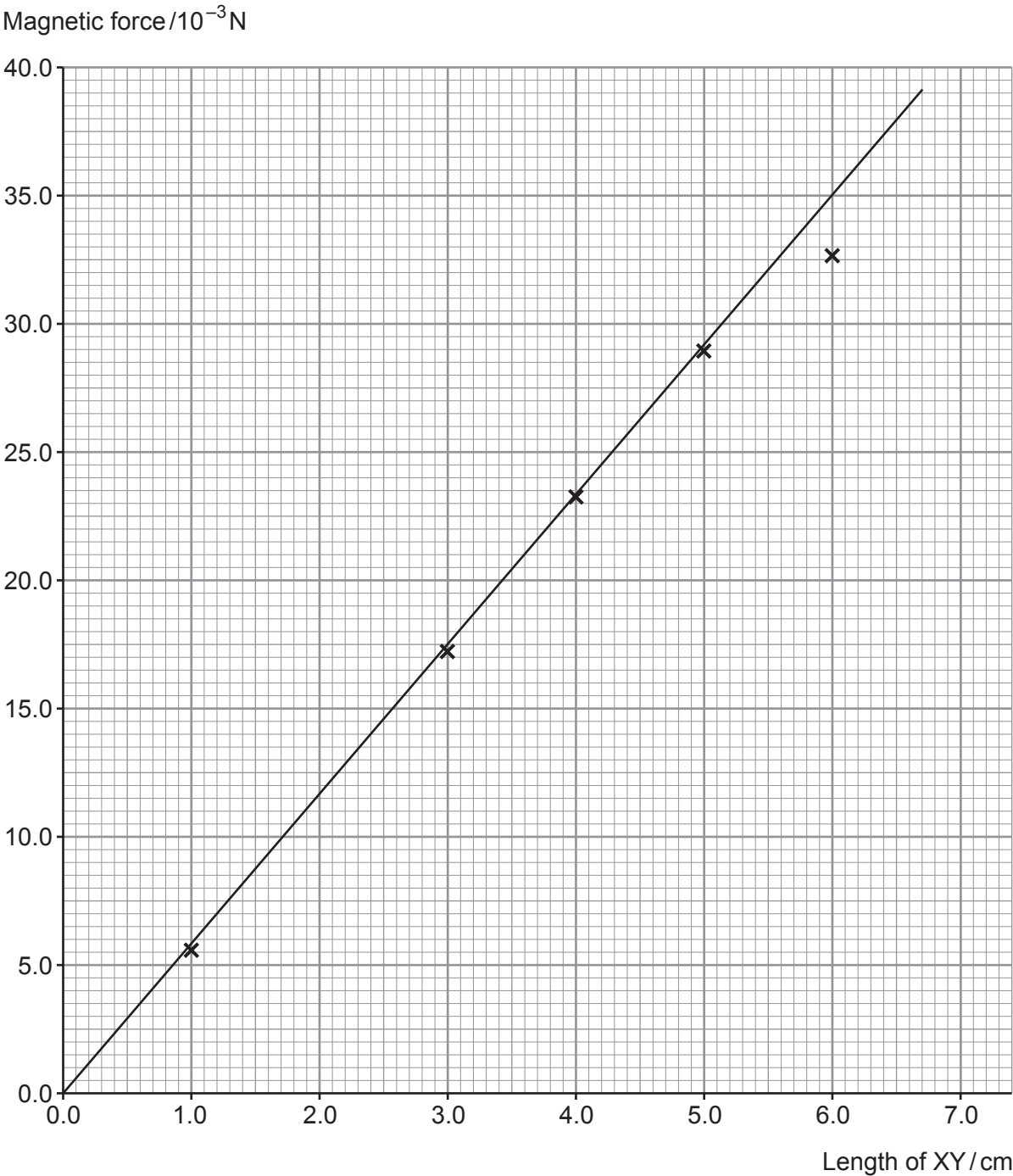


- (b) Bronwen varies the length of XY and records the electronic balance reading each time when a current of 5.00 A flows. She records all her results in a table and plots a graph of magnetic force against length of wire.
- (i) **Complete the table and plot** the two missing points on the graph. [4]

Length of XY/cm	Balance reading/g	Magnetic force/10 ⁻³ N
1.0	0.56	5.5
2.0	1.22	
3.0	1.75	17.2
4.0	2.36	23.1
5.0	2.89	28.4
6.0	3.32	32.6
7.0		35.8



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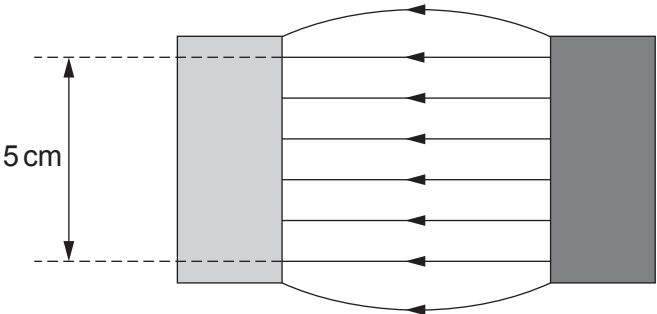


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- (ii) The region of uniform magnetic field between the poles of the magnet is of length 5 cm only. Evaluate whether or not Bronwen has drawn the line of best fit correctly. [3]



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- (iii) Determine the magnetic flux density, B , of the uniform magnetic field between the poles of the magnet and quote it to an appropriate number of significant figures. The current in the wire is 5.00 A. No uncertainty is required in your value of B . [5]

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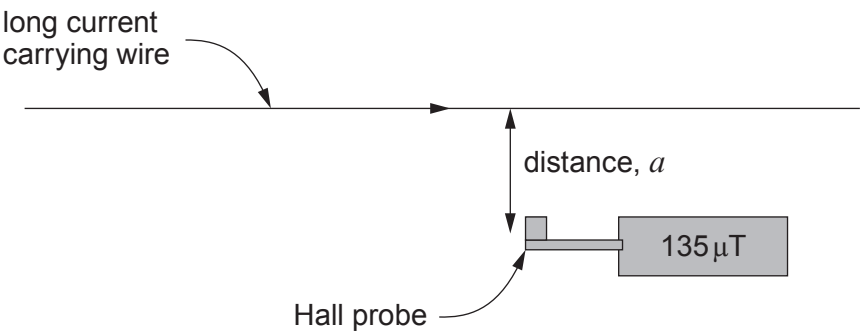
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5. Michael carries out an experiment to determine how the magnetic field, B , due to a long current carrying wire varies with distance, a , from the wire.



His results are recorded in a table and a graph of $\ln(B)$ against $\ln(a)$ is plotted using the corrected B -field.

a/m	Measured B -field / T	Corrected B -field, B/T	$\ln(a/\text{m})$	$\ln(B/\text{T})$
0.010	0.000 529	0.000 499	-4.61	-7.60
0.020	0.000 279	0.000 249	-3.91	-8.30
0.040	0.000 157			
0.080	0.000 092	0.000 062	-2.53	-9.69
0.160	0.000 063	0.000 033	-1.83	-10.32
0.300	0.000 046	0.000 016	-1.20	-11.04

- (a) Explain why the corrected B -field, B must be obtained. [1]

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- (b) Complete the table above and plot the missing point on the graph opposite (no error bars are required). [3]

- (c) The gradient of the steepest line is -1.03 . Calculate the gradient of the least steep line. [2]

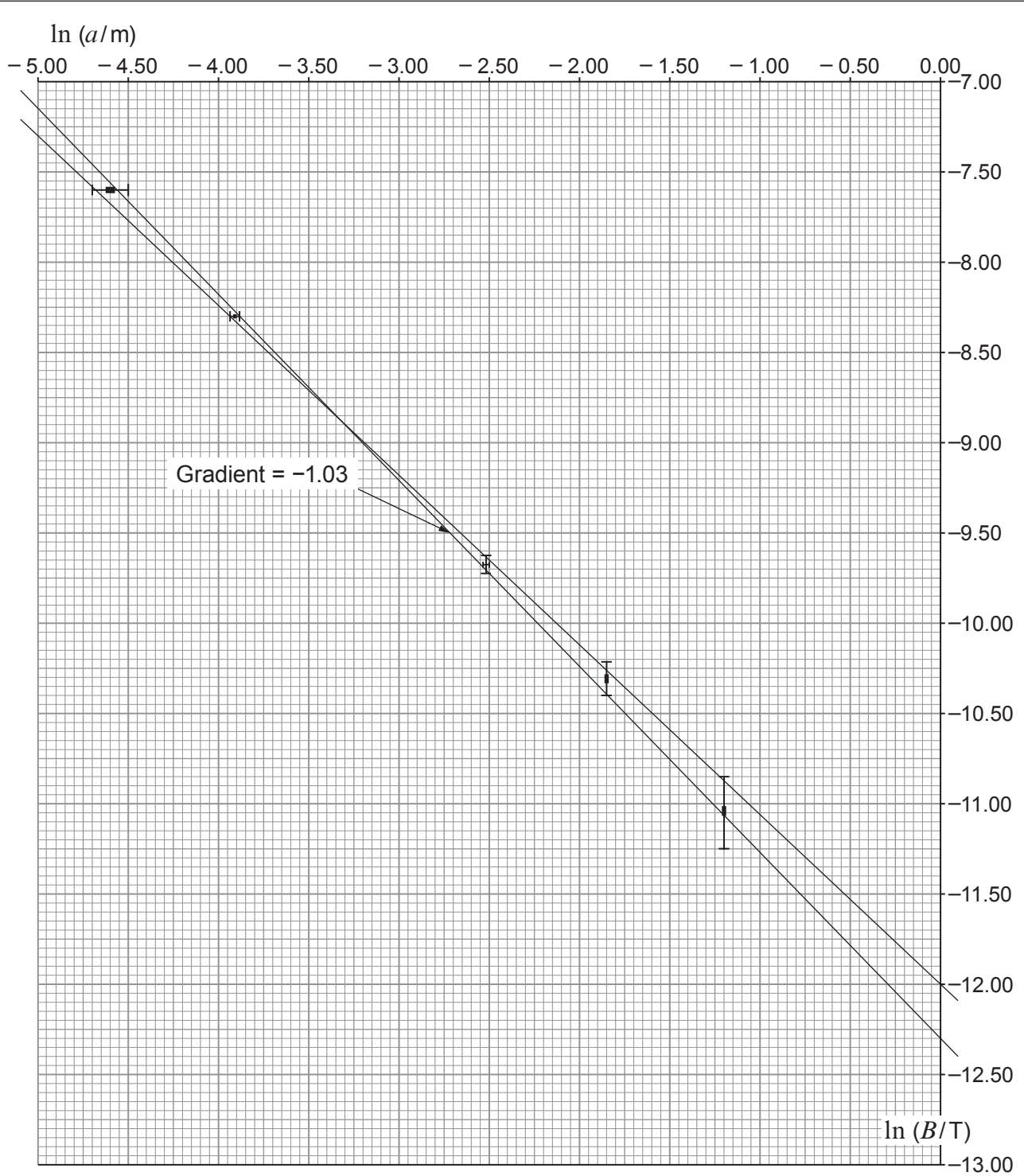
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(d) Use the equation $B = \frac{\mu_0 I}{2\pi a}$ to show that the expected mean gradient is -1 and that the expected $\ln(B)$ intercept is $\ln\left(\frac{\mu_0 I}{2\pi}\right)$. [2]

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(e) Explain to what extent the graph confirms the relationship $B = \frac{\mu_0 I}{2\pi a}$. [3]

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(f) Calculate the current in the wire along with its **absolute** uncertainty. [5]

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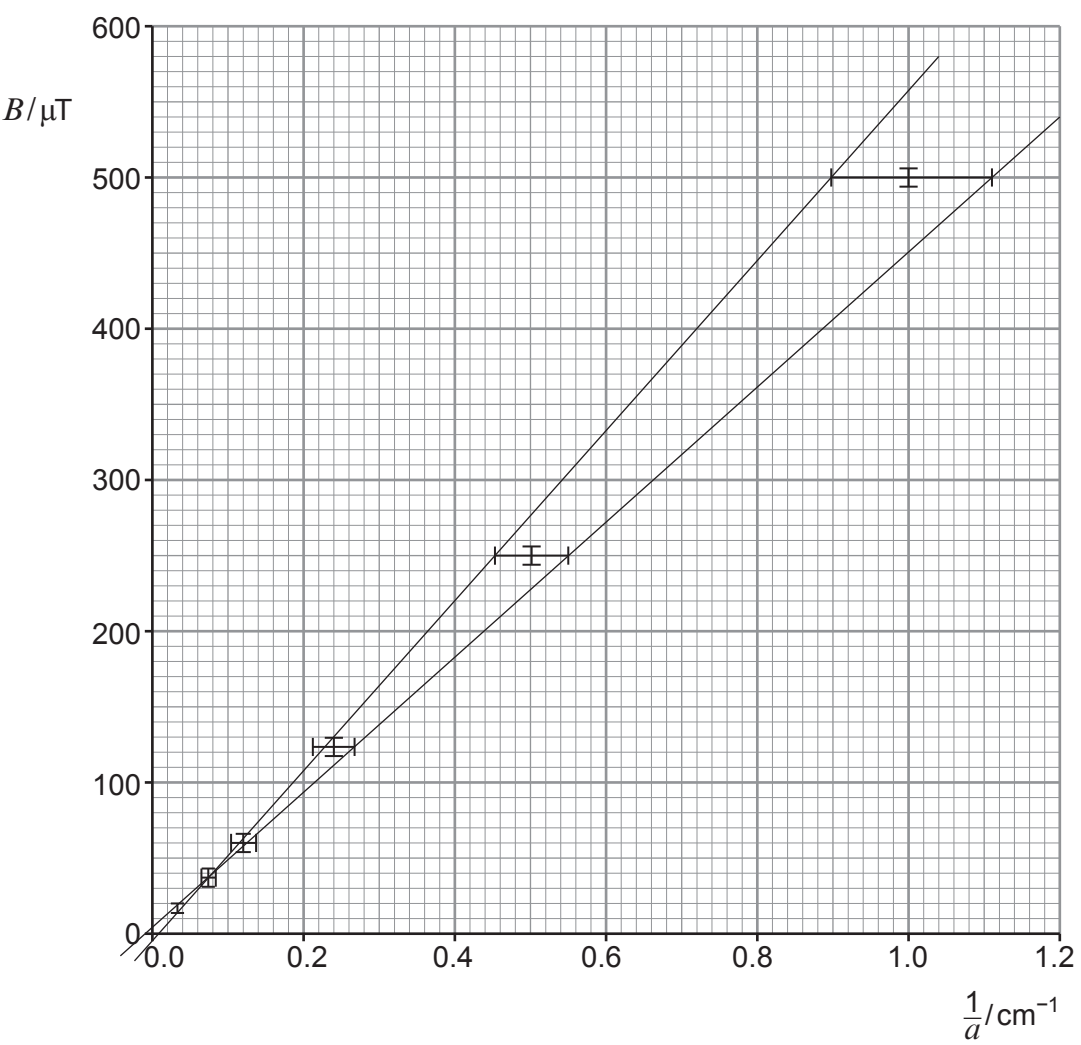
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(g) Bethan analyses Michael's data by plotting a graph of B against $\frac{1}{a}$.



Explain briefly why Michael's log graph is more suitable than Bethan's graph for this data set (no calculations are required). [2]

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